Original LP is "overly precise": near-zero regret even with wrong λ
Aggregation makes the model coarser ⇒ regret grows faster with parameter error

Regret $V^*(\lambda_{true}) - V_{true}(\pi(\lambda))$

Original LP (S=56)
Random W (M=10)
Random W (M=20)
Random W (M=30)
Trained W (M=10)
Trained W (M=20)
Trained W (M=30)
$\lambda_{true} = 10.0$
2% of V^*

DFL gradient signal: original LP has near-zero $dV/d\lambda$ (no learning signal)
Aggregation amplifies gradient and points toward λ_{true}

$dV_{true}/d\lambda$

Original LP (S=56) [near-zero: 67%]
Random W (M=10) [near-zero: 29%]
Random W (M=20) [near-zero: 5%]
Random W (M=30) [near-zero: 5%]
Trained W (M=10) [near-zero: 37%]
Trained W (M=20) [near-zero: 37%]
Trained W (M=30) [near-zero: 43%]

Mechanism: original LP finds similar policies for different λ (robust)
Aggregated LP policies diverge faster (sensitive to parameter error)

Policy distance $\|y(\lambda) - y(\lambda_{true})\|_1$

λ_{pred}

Original LP (S=56)
Random W (M=10)
Random W (M=20)
Random W (M=30)
Trained W (M=10)
Trained W (M=20)
Trained W (M=30)